

Response to the Editors and Reviewers

IEEE Transactions on Circuits and Systems for Video Technology

Manuscript TCSVT-29693-2026: *Beyond Homogeneous Adaptation: Decoupled Control for Sample- and Layer-Aware Test-Time Robustness in Temporally Correlated Streams*

Dear Editor-in-Chief, Senior Area Editor, Associate Editor, and Reviewers,

Thank you for your careful evaluation and constructive suggestions. We appreciate your recognition of the motivation and long-horizon evidence of the Decoupled Control Framework (DCF). The revision has sharpened the central contribution of DCF and strengthened its analytical, experimental, and computational validation.

In response to the Editor-in-Chief’s guidance, we identify four closely related TCSVT studies—QED [31], MetaBN [29], CMDA [30], and MS-TTA [36]—and clarify DCF’s methodological advances beyond their respective approaches. Rather than treating sample reliability and model stability as separate concerns, DCF formulates their interaction as a coupled sample–layer feedback problem and introduces a unified route–adapt–retain framework to control it. Specifically, PSR differentiates supervisory roles, RGR recovers geometric information from routed-away samples, and CLR regulates the persistence of candidate updates across network depth. Their coordinated operation addresses the accumulation of adaptation errors in long, temporally correlated streams. Detailed comparisons and supporting evidence are provided in our response to the Editor-in-Chief and Sections I–III of the revised manuscript.

Summary of major technical and experimental revisions.

1. Probe interpretation and layer behavior. We strengthened the local directional-sensitivity interpretation of the Perturbation Consistency Score (PCS) through analytical expansion and added controlled shape–texture and Colored-MNIST experiments, matched-coverage routing diagnostics, and frequency/strength sensitivity analyses (Tables IV–VI, Figs. 5 and 12(c)). For Curvature-aware Layer Retention (CLR), we added retention-gate heatmaps over time and corruption types, compared control granularities, and evaluated the default gradient-square proxy against expected Fisher and a Hutchinson-based second-order approximation (Fig. 9, Table IX).

2. Geometry repair and computational cost. We compared Routed-away Geometry Repair (RGR) with moment matching, prototype contrastive alignment, and uniform-prior optimal transport, and examined routed-away proportions from 10% to 90%. These tests identify a stable operating regime and a failure regime when too little trusted evidence remains (Tables VII–VIII). We also benchmarked FLOPs, peak memory, and latency alongside representative baselines. On an NVIDIA A100, DCF-Lite achieves 42.06% long-horizon average accuracy with 61.8% fewer FLOPs and 58.6% lower peak memory than full DCF (Table XI).

3. Empirical rigor and reproducibility. We added DomainNet-126 as a distinct multi-domain transfer benchmark. Across five matched runs, paired t -tests establish significant aggregate transfer improvements over the strongest aggregate baseline on each benchmark (Fig. 11). We also incorporated the suggested sensor-stream literature ([19]–[21]) and recent continual-TTA studies ([28], [38]), and clarified trainable parameters, CLR block grouping, routing regions, threshold selection, and Sinkhorn OT settings.

Below, we respond to each editorial request and reviewer comment in detail and identify the corresponding manuscript revisions. We respectfully submit the revised manuscript for your further consideration.

Sincerely,

Yuanyuan Pu

Corresponding author, on behalf of all authors

Roadmap of Responses to Editors and Reviewers

Editor / Reviewer Group	Direct Jump
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Reviewer 2 (R2)	Jump → p. 19

Response to the Editor-in-Chief

Related TCSVT studies and the distinctive contribution of DCF

Before your new submission, you have to answer two questions clearly in your revised manuscript and responses: a) what are the 3-5 papers published in the IEEE Transactions on Circuits and Systems for Video Technology, which are most closely related to your manuscript; b) what is distinctive / new about your current manuscript related to these previously published papers.

Online response (EIC) [↗](#)

PDF diff [↗](#)

► Response:

We appreciate the Editor-in-Chief’s request to establish both the closest connections to prior TCSVT work and the substantive novelty of DCF. We answer the two questions separately. Rather than distinguishing DCF through implementation choices alone, we clarify the control problem it formulates, the online decisions it introduces, and the evidence that these decisions matter during prolonged adaptation.

1. Response to Question a): Closely Related Papers Published in IEEE TCSVT

In the revised Section II (“Related Work”), we have explicitly incorporated and analyzed four closely related studies recently published in IEEE TCSVT (spanning 2024 to 2026) that address test-time adaptation, target-evidence reliability, online model adaptation, or target-side representation refinement:

- 1) **QED** (Liu *et al.*, IEEE TCSVT 2024) [31]: *Question Type-Aware Debiasing for Test-Time Visual Question Answering Model Adaptation*.
Focus & Mechanism: QED addresses test-time bias in VQA by leveraging question-type-aware entropy and negative samples to identify predictions likely to be influenced by language priors. Its primary control determines whether a test instance should contribute to adaptation according to task-specific question–answer structure.
- 2) **MetaBN** (Liu and Yang, IEEE TCSVT 2025) [29]: *Test-Time Adaptation for Real-World Video Adverse Weather Restoration With Meta Batch Normalization*.
Focus & Mechanism: MetaBN develops adaptation capability through source-stage meta-learning of normalization affine parameters and applies self-supervised updates online for adverse-weather video restoration. Its adaptation mechanism is therefore closely tied to source-stage preparation of normalization behavior.
- 3) **CMDA** (Zhang *et al.*, IEEE TCSVT 2026) [30]: *Collaborative Model and Data Adaptation at Test Time*.
Focus & Mechanism: CMDA couples predictive model adaptation with diffusion-based data refinement in a bidirectional loop: current predictions guide reverse-diffusion sample generation, and the generated samples subsequently refine the predictive model.
- 4) **MS-TTA** (Han *et al.*, IEEE TCSVT 2026) [36]: *Unleashing the Potential of All Test Samples: Mean-Shift Guided Test-Time Adaptation*.
Focus & Mechanism: MS-TTA exploits information from test samples without conventional online parameter optimization by applying mean-shift refinement to CLIP embeddings and maintaining a feature cache during inference.

2. Response to Question b): Distinctive Novelty and Advances of DCF

Core contribution: DCF makes the evidence–update–retention feedback of long-horizon TTA an explicit target of online control. In temporally correlated streams, model predictions determine which target observations appear suitable for self-supervision; those observations induce parameter updates; and the retained updates change the predictions and supervision available at subsequent steps. This feedback can amplify errors when the reliability of individual samples and the persistence of updates across network depth are not controlled together. DCF formulates this interaction as a coupled sample–layer feedback problem, rather than treating adaptation as uniform confidence-based self-training or a single global update/regularization decision.

The resulting **route–adapt–retain architecture** makes three complementary decisions within every online adaptation step. **First**, Probe-supported Sample Routing (PSR) combines predictive entropy and Fourier-based Perturbation Consistency Score (PCS) to assign distinct supervisory roles, distinguishing trusted samples from confident but stress-inert candidates. **Second**, Routed-away Geometry Repair (RGR) does not simply discard samples that lack sufficient reliability for direct supervision: it uses their representations in source-anchored, dynamic-prior optimal-transport constraints, thereby extracting structural

information without treating their predicted labels as equally trustworthy. **Third**, Curvature-aware Layer Retention (CLR) applies a source–candidate curvature-mismatch criterion after candidate update generation to determine, by layer group, how much of an update should persist. These decisions are linked: evidence routing shapes the candidate update, geometry repair uses the other part of the stream, and retention regulates how that update influences future evidence assessment. **This explicit linkage between sample-level evidence roles and depth-dependent parameter persistence is the principal methodological distinction of DCF.**

The advance relative to the four related TCSVT studies is as follows:

- **Relative to QED [31]: From bias-aware evidence screening to differentiated evidence utilization.** QED addresses language-prior bias in test-time visual question answering through question-type-aware entropy and negative-sample mechanisms. DCF addresses a different reliability problem in temporally correlated recognition streams. PSR uses complementary confidence and structured stress responsiveness to identify which samples can directly support online adaptation. Its key distinction is not merely an additional confidence criterion: the routing decision changes how different samples contribute. Trusted observations provide direct consistency supervision, while routed-away observations supply geometry constraints through RGR. This creates a two-role evidence-control mechanism within a sustained online update process, rather than reducing unreliable evidence to a binary accept/discard decision.
- **Relative to MetaBN [29]: From prepared normalization adaptability to explicit, depth-specific update persistence.** MetaBN builds test-time weather-restoration adaptability through source-stage meta-learning of normalization affine parameters followed by online self-supervised updates. DCF instead introduces a post-update retention decision that directly controls the parameter state carried forward. CLR measures source–candidate curvature mismatch across layer groups and computes a proxy-weighted combination of source and candidate parameters, allowing some groups to remain plastic while others are anchored more strongly to the source model. Thus DCF controls not only how an update is formed but where it survives, without requiring source-stage meta-training. This matters specifically when prolonged adaptation produces different stability risks across network depth.
- **Relative to CMDA [30]: From model–data co-refinement to feedback control over evidence and persistent model state.** CMDA develops a bidirectional feedback loop between predictions and diffusion-based data refinement. DCF focuses on a distinct feedback mechanism: current model predictions influence which observed samples provide direct supervision, and the resulting retained parameter updates shape later predictions and sample selection. The route–adapt–retain design regulates this mechanism at both ends. PSR and RGR control the contribution of heterogeneous target evidence before and during update generation, while CLR explicitly controls the layer-wise persistence of the candidate update. CMDA emphasizes model–data co-refinement, whereas DCF regulates evidence utilization and the persistence of parameter updates throughout online adaptation.
- **Relative to MS-TTA [36]: From training-free representation refinement to geometry-informed, selectively retained parameter adaptation.** MS-TTA exploits test samples through mean-shift refinement of CLIP representations and an inference-time feature cache, without conventional online parameter optimization. DCF also seeks to avoid wasting information from difficult samples, but gives them a different role. RGR uses soft OT assignments to source-initialized centroids, with a conservative dynamic class marginal, to generate a geometry objective that participates in online parameter learning. CLR then determines which parts of that learned change persist across layer groups. Hence the distinction is not a general claim to be the first to reuse uncertain samples; it is the integration of their geometric contribution into an online update whose long-term persistence is itself controlled.

Why the integration is substantive rather than additive. The three modules regulate successive, mutually dependent parts of the same adaptation process. PSR determines who may supply direct supervisory evidence; RGR preserves a constrained use for samples excluded from that channel; and CLR prevents the entire resulting candidate update from being propagated uniformly through the model. Removing the retention decision leaves sample-side routing and repair unable to control how an inaccurate update influences later batches. Conversely, retention alone does not determine which observations should have been trusted or how the remainder should be used. DCF therefore contributes a framework for controlling the flow and persistence of adaptation information, not merely three unrelated enhancements.

Mechanism-level and long-horizon evidence. The controlled ablation in Table III directly tests the complementary role of sample-side and layer-side control: PSR+RGR yields 26.25% final accuracy at Round 15, whereas adding CLR increases that result to 43.84% (+17.59 pp). Leave-one-out variants further isolate the roles of PSR and RGR. Tables IV–VI support probe-based routing with shortcut and matched-coverage diagnostics; Tables VII–VIII compare geometry-repair alternatives and identify routing-ratio limits; and Fig. 9 and Table IX characterize the temporal and layer-wise behavior of the retention mechanism. Table II reports the strongest long-horizon average accuracy among the evaluated baselines under T-CS, T-CS-LS, and T-CS-MSL on the 225-domain stream, and Fig. 11 tests transfer beyond the principal benchmark. These findings support

the claimed benefits of coordinated evidence assignment and update retention.

Changes in the manuscript: Section I (p. 2); Section II (p. 3); Section III-B (p. 3, Eqs. (2)–(3)); Section IV-B & IV-C (pp. 7–12, Tables I–XI; Figs. 9, 11, 12); References [29], [30], [31], [36] (p. 13).

Selected Revised Text in Section II (Page 3):

“Related studies further explore question-conditioned debiasing [31], meta-trained normalization for adverse-weather adaptation [29], diffusion-based collaborative model–data adaptation [30], and training-free mean-shift refinement of test representations [36]. In contrast, DCF controls the coupled evolution of target evidence and model parameters in temporally correlated recognition streams. PSR routes evidence by entropy and Fourier stress responsiveness; RGR converts routed-away samples into soft OT geometry constraints; and CLR determines where the resulting candidate update persists.”

“DCF formulates online adaptation as a coupled sample–layer feedback process, jointly controlling which evidence drives direct supervision, how routed-away samples constrain target geometry, and where candidate updates persist. This route–adapt–retain design interrupts the error-amplifying feedback that causes long-horizon collapse.”

Response to the Senior Area Editor

Relevance, completeness, and ablation depth

The paper proposes the use of a decoupled control system for reducing bias in target evidence in test-time adaptation using online methods for temporally correlated image streams. The topic is relevant to TCSVT. The paper is complete with contributions, experiments, results, and a limited ablation study.

 [Online response \(SAE\) ↗](#)

 [PDF diff ↗](#)

► Response:

Thank you for recognizing the relevance and completeness of the work. We expanded the ablation study in Section IV-C with controlled diagnostics for DCF's Probe-supported Sample Routing (PSR) module; alternatives, prior ablation, and routing-ratio stress tests for DCF's Routed-away Geometry Repair (RGR) module; gate visualizations, granularity and proxy comparisons for DCF's Curvature-aware Layer Retention (CLR) module; cross-domain statistics; and DCF-Lite efficiency analysis. These additions establish the contribution of each control mechanism, reveal their interactions, and quantify the accuracy–efficiency trade-off.

Changes in the manuscript: Section IV-C, Tables IV–IX and XI, pp. 8–12; Figs. 9, 11, and 12, pp. 10–12.

Response to the Associate Editor

Thank you for identifying the analyses needed for this major revision. We address the five substantive concerns below; the corresponding Reviewer 1 responses provide the detailed evidence.

Associate Editor: Comment Overview and Navigation

Item	Comment Title	Direct Jump
AE1	Justification and sensitivity of the Fourier probe	AE1 (p. 7) → [R1.1]
AE2	Alternatives and failure cases for geometry repair	AE2 (p. 7) → [R1.2]
AE3	Computational cost and practical approximations	AE3 (p. 8) → [R1.3]
AE4	Behavior of the layer retention gates	AE4 (p. 8) → [R1.4]
AE5	Statistical evidence for cross-domain transfer	AE5 (p. 8) → [R1.6]

AE1: Justification and sensitivity of the Fourier probe

The Fourier-based stress probe lacks formal theoretical justification, and a sensitivity study on the choice of frequency range and perturbation strength is needed.

[Online response \(AE1\) ↗](#)

[PDF diff ↗](#)

► Response:

We now ground the Fourier probe in a local directional-sensitivity analysis and test its shortcut sensitivity directly with controlled experiments. The Perturbation Consistency Score (PCS), used by DCF’s Probe-supported Sample Routing (PSR) module, is the positive drop in the original predicted-class probability, and the expansion in Response R1.1 explicitly retains this positive-part operation. The analysis identifies what PCS measures—one-sided sensitivity along the structured Fourier directions—while the shape–texture interventions, paired shortcut counterfactuals, matched-coverage ImageNet-C purity analysis, and Colored-MNIST jointly establish that probe responsiveness provides discriminative, shortcut-sensitive routing information beyond confidence alone. Figure 12(c) further demonstrates stable adaptation accuracy across the evaluated frequency ranges and perturbation strengths.

Changes in the manuscript: Section III-C, p. 4; Section IV-C, Tables IV–VI, pp. 8–9, and Fig. 12(c), p. 12. See Response R1.1.

AE2: Alternatives and failure cases for geometry repair

The routed geometry repair module should be compared against simpler alternatives, and potential failure cases when the routed set is small or the prior estimate is noisy should be discussed.

[Online response \(AE2\) ↗](#)

[PDF diff ↗](#)

► Response:

We added the requested alternatives and tested how DCF’s Routed-away Geometry Repair (RGR) module behaves as the proportion of routed-away samples changes. Table VII compares discarding routed-away samples, feature moment matching, prototype contrastive alignment, fixed-uniform-prior OT, and the proposed dynamic-prior OT while keeping the remaining DCF components unchanged. We further added a routing-ratio stress test from 10% to 90% routed-away samples and clarified the prior EMA, uniform interpolation, centroid momentum, and empty-set safeguards. These tests show stable performance when enough trusted samples remain, and a clear decline when routed-away samples dominate the batch.

Changes in the manuscript: Section III-D, pp. 4–5; Section IV-C, Tables VII–VIII, p. 10; Algorithm 1, p. 6. See Response R1.2.

AE3: Computational cost and practical approximations

Computational overhead, including FLOPs and memory, should be reported in detail alongside a discussion of possible approximations for practical deployment.

[Online response \(AE3\)](#)
[PDF diff](#)
► Response:

Table XI compares FLOPs, peak GPU memory, per-image runtime, and long-horizon average accuracy for DCF and representative baselines under the same A100 evaluation setting. Full DCF achieves 43.48% long-horizon average accuracy with 24.6 GFLOPs, 14,690 MB peak memory, and 5.20 ms/image. DCF-Lite reduces the cost to 9.4 GFLOPs and 6,080 MB with 3.19 ms/image while retaining 42.06% accuracy, corresponding to reductions of 61.8% in FLOPs and 58.6% in peak memory at the cost of only a 1.42 pp drop in accuracy. Figure 12(d) supports using few Sinkhorn iterations, while Table IX quantifies the additional cost of Expected Fisher and Hutchinson-based estimators and supports the efficient default gradient-square proxy.

Changes in the manuscript: Section IV-C, “Computational Overhead,” pp. 11–12; Table XI and Fig. 12(d), p. 12. See Response R1.3.

AE4: Behavior of the layer retention gates

The per-layer behaviour of the curvature-aware retention mechanism needs further analysis, for example through a heatmap of retention gates over time and corruption types.

[Online response \(AE4\)](#)
[PDF diff](#)
► Response:

Figure 9 now shows the layer-wise retention gates of DCF’s Curvature-aware Layer Retention (CLR) module throughout the 225-domain stream and across corruption types. The gates vary with network depth and corruption type: layer4 maintains the largest retention gates, while earlier and intermediate groups increase source anchoring temporarily in response to specific corruptions. The temporal trajectories recover after transient suppression, demonstrating that CLR preserves adaptation capacity through dynamic retention. Table IX further compares control granularities and alternative curvature estimators.

Changes in the manuscript: Section III-E, pp. 5–6; Section IV-C, Fig. 9 and Table IX, p. 10. See Response R1.4.

AE5: Statistical evidence for cross-domain transfer

Additionally, statistical significance should be reported for the cross-domain results, where some improvements are marginal.

[Online response \(AE5\)](#)
[PDF diff](#)
► Response:

Figure 11 now reports mean $\pm$ standard deviation over five matched runs. The accompanying cross-domain analysis uses paired tests against the strongest aggregate baseline for each benchmark. On ImageNet-C, DCF achieves a mean gain over No Adapt of 30.30 ± 0.39 pp versus 27.95 ± 0.05 pp for AEA, with $t(4) = 14.30$ and $p = 1.39 \times 10^{-4}$. On DomainNet-126, DCF achieves 61.40% mean accuracy versus 60.13% for DeYO, with $t(4) = 12.01$ and $p = 2.75 \times 10^{-4}$. These paired tests establish statistically significant aggregate differences across matched runs; individual transfer cells are reported with their run-to-run variability.

Changes in the manuscript: Section IV-C, “Cross-Domain Transfer,” and Fig. 11, p. 11. See Response R1.6.

Response to Reviewer 1

We appreciate your positive assessment of the motivation and long-horizon results. Your comments motivated additional analytical grounding, controlled shortcut experiments, mechanism-level robustness studies, and deployment-oriented evaluation. We address the six comments individually below.

Reviewer 1: Comment Overview and Navigation

Item	Comment Title	Direct Jump
R1.1	Fourier probe justification and sensitivity	Jump → p. 9
R1.2	Geometry repair alternatives and failure modes	Jump → p. 11
R1.3	Computational overhead and lightweight deployment	Jump → p. 13
R1.4	Layer retention patterns and alternative proxies	Jump → p. 14
R1.5	Additional literature on uncertainty-aware modeling	Jump → p. 16
R1.6	Variability, significance, and transfer beyond ImageNet	Jump → p. 16

R1.1: Fourier probe justification and sensitivity

The paper empirically shows that the Fourier perturbation exposes stress-inert predictions, but lacks a formal justification of why frequency-domain stress specifically targets shortcut cues (e.g., texture biases) rather than semantic features. A theoretical analysis—or at least a controlled synthetic experiment where shortcuts are known (e.g., color vs. shape)—would strengthen the claim. Currently, the probe design (random frequency and direction) appears heuristic; a sensitivity study on the choice of frequency range and perturbation strength λ would also help practitioners set these parameters without extensive tuning.

 [Online response \(R1.1\)](#)

 [PDF diff](#)

► Response:

Thank you for this suggestion. We added a local sensitivity analysis, controlled experiments with known shortcut factors, matched-coverage routing diagnostics, and a broader sensitivity study. Together, these additions provide both an analytical interpretation of the Perturbation Consistency Score (PCS) used by DCF’s Probe-supported Sample Routing (PSR) module and direct evidence that its stress responsiveness contributes reliable routing information beyond predictive confidence alone.

Local squared-response interpretation of PCS

Section III-C defines PCS as the positive drop in the probability assigned to the original predicted class under the stressed view. For a fixed input x and its original predicted class, let q denote the distribution induced by the probe sampling and clamping, $\delta = \mathcal{A}^s(x) - x$ the resulting perturbation, and g_x the input gradient of the original predicted-class probability. Under local smoothness and sufficiently small perturbations, the expansion below characterizes the expected squared PCS. To make the positive-part operation in Eq. (5) explicit, the local squared-response expansion is

$$\mathbb{E}_q[s(x)^2] = \mathbb{E}_q\left[\left(-g_x^\top \delta\right)_+^2\right] + O(\mathbb{E}_q\|\delta\|_2^3).$$

Thus PCS measures one-sided probability sensitivity along the directions excited by the Fourier probe. Let $Q_q(x) = \mathbb{E}_q[\delta\delta^\top]$ denote the perturbation second-moment matrix. The untruncated quadratic response is $g_x^\top Q_q(x)g_x$; for the positive-part PCS, the leading term retains the truncation and is bounded above by this quadratic response. The relation uses the actual post-clamp perturbation second moment and accommodates asymmetric perturbations.

The local analysis and controlled experiments provide complementary evidence for the probe design:

- **Analytical role of the local expansion:** Under local smoothness and sufficiently small perturbations, the leading term of the expected squared PCS is $\mathbb{E}_q[(-g_x^\top \delta)_+^2]$, which is bounded above by $g_x^\top Q_q(x)g_x$. This characterizes PCS as a one-sided directional-sensitivity signal under the Fourier probe distribution and provides a precise analytical basis for interpreting its response.
- **Empirical basis for shortcut-sensitive routing:** The controlled shape–texture experiment directly tests the relationship between probe response and visual factors. The random Fourier probe yields mean PCS values of 229.0×10^{-3} for

the shape readout and 64.0×10^{-3} for the texture readout, while preserving the original shape decision in 99.60% of cases (Table IV). These metrics capture two aspects of the response: the probability assigned to the original shape class changes under stress, while the predicted shape class remains highly consistent. The larger shape PCS therefore provides a factor-sensitive response for routing. More directly, the paired counterfactual analysis distinguishes task-relevant from shortcut-driven predictions among confidence-matched candidates with an AUROC of 0.81, compared with 0.71 for energy-matched pixel noise. These results establish discriminative routing value in the controlled setting, complementing the directional-sensitivity analysis.

- **Threshold selection:** The default $v_{\text{PCS}} = 0.20$ is evaluated through the routing-threshold sensitivity study in Fig. 12(b). Table IV provides complementary evidence of differential factor sensitivity; the routing benefit is assessed directly by the AUROC and matched-coverage diagnostics. Together, these analyses support the default threshold and show that PCS improves selection among confident candidates.

Controlled shortcut evidence

We added a 224×224 synthetic benchmark in which circle/square shape is the task factor and four sinusoidal textures provide explicitly controlled shortcuts. Table IV reports agreement between the original and stressed shape predictions together with PCS from the shape and texture readouts. These measurements characterize label consistency and factor-dependent probability sensitivity.

Table IV. Factorized response diagnostics of the Fourier probe. Shape agreement measures consistency between original and stressed shape predictions, while PCS measures the response of the corresponding factor-specific readout. The shape/texture PCS ratio characterizes the probe’s differential sensitivity across visual factors. Lower, middle, and upper nominal bands correspond to $[1, 75)$, $[75, 150)$, and $[150, 224]$ cycles/image, respectively.

Probe	Shape agreement (%)	Shape PCS ¹	Texture PCS ¹	PCS ratio
Lower nominal band	99.40	241.0	69.0	3.49×
Middle nominal band	99.60	224.0	62.0	3.61×
Upper nominal band	99.70	217.0	58.0	3.74×
Energy-matched pixel noise	99.80	112.0	98.0	1.14×
Random Fourier probe	99.60	229.0	64.0	3.58×

¹ PCS entries are in units of 10^{-3} . The three nominal bands are $[1, 75)$, $[75, 150)$, and $[150, 224]$ cycles/image. “Nominal” frequencies parameterize the probe before grid sampling, which can introduce aliasing.

Across preserved, randomized, and reversed shortcut correlations, PCS achieves 0.81 AUROC for distinguishing task-relevant from shortcut-driven predictions, compared with 0.71 for energy-matched pixel noise. At matched trusted-set coverage, shortcut contamination decreases from 24.6% to 11.8%. On ImageNet-C, Table VI further compares filters at the same frozen model state and matched coverage: pseudo-label purity is 49.9% for entropy + PCS, versus 44.0% for entropy alone and 47.1% for PCS alone. At matched coverage, joint entropy–PCS routing improves pseudo-label purity by 5.9 pp over entropy-only selection and by 2.8 pp over PCS-only selection. Together with the confidence-matched counterfactual results, these gains establish the complementary routing value of confidence and structured stress responsiveness.

Table V evaluates the complete DCF framework on binary Colored-MNIST, where digit identity defines the semantic target and color provides a controlled shortcut. Training labels contain 25% noise, while test labels are clean and color–label agreement is approximately 30%. DCF achieves 88.91% accuracy, exceeding the strongest evaluated baseline, DeYO, by 26.54 pp and substantially improving on the source model’s 50.93%.

Table V. Controlled color shortcut on binary Colored-MNIST. Digit identity defines the semantic label, while color serves as a controlled spurious cue. Training labels contain 25% noise, whereas test labels are clean, with approximately 30% color–label agreement.

Method	Source	BN	Tent	CoTTA	RoTTA	SAR	DeYO
Acc. (%)	50.93	47.21	44.36	47.21	50.23	47.21	62.37
Method	AEA	TRIBE	LAW	SPA	PTTA	DCF (Ours)	
Acc. (%)	43.18	33.93	42.73	39.26	50.81	88.91	

Table VI. Selected-sample quality on ImageNet-C (T-CS) at matched trusted-set coverage. Pseudo-label purity measures correctness of selected pseudo-labels.

Method	Pseudo-label Purity (%)
Tent	42.7
+ Entropy Filter	44.0
+ PCS Filter	47.1
+ Entropy & PCS	49.9

Frequency range and perturbation strength

Figure 12(c) jointly varies the nominal frequency sampling range and perturbation strength λ . Section III-C specifies independent per-channel frequency and direction sampling, RMS-normalized Fourier bases, and exponential amplitude sampling with rate $1/\lambda$; the default is $\lambda = 0.2$ with nominal frequencies sampled from $[1, 224]$ cycles/image. Performance remains stable across a broad parameter region around this configuration, supporting the reported setting as a practical default.

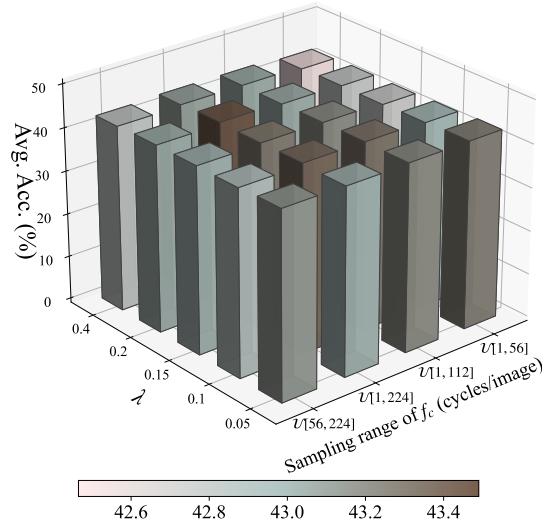


Fig. 12(c). Hyperparameter sensitivity. Sensitivity to perturbation strength λ and nominal frequency sampling range f_c (cycles/image).

Changes in the manuscript: Section III-C, p. 4, Eqs. (4)–(7); Section IV-C, “Analysis of Probe-Supported Sample Routing,” pp. 7–9; Tables IV–VI, pp. 8–9; Fig. 12(c), p. 12.

R1.2: Geometry repair alternatives and failure modes

RGR uses entropic OT with dynamic class priors to align routed-away features to source-initialized centroids. However, the paper does not compare this with simpler alternatives like feature moment matching or contrastive learning. Moreover, the centroid update (Eq. 19) uses only routed-away samples; if the routed set dominates, centroids may drift away from source semantics. The authors should discuss failure cases when the routed set is small or when the prior estimate is noisy (e.g., early adaptation stages). Adding an ablation with fixed uniform priors would isolate the benefit of dynamic priors.

Online response (R1.2) [↗](#)

PDF diff [↗](#)

► Response:

Thank you for suggesting these comparisons. We added them to separate the effects of OT-based geometry repair and the dynamic prior, and used a routing-ratio stress test to examine when DCF’s Routed-away Geometry Repair (RGR) module remains effective.

Comparison with simpler alternatives

Table VII keeps DCF’s Probe-supported Sample Routing (PSR) and Curvature-aware Layer Retention (CLR) modules fixed and changes only the treatment of the routed-away subset. It compares discarding those samples, feature moment matching, prototype contrastive alignment, OT with a fixed uniform prior, and OT with the proposed dynamic prior.

Table VII. Comparison of alternative treatments for routed-away samples on ImageNet-C. All variants share the same PSR and CLR.

Treatment of routed-away samples	T-CS (%)	T-CS-LS (%)	Long-horizon final (%)	Avg. (%)
Discard samples	41.87	42.71	42.88	42.49
Moment matching	42.81	42.82	41.51	42.38
Prototype contrastive alignment	42.83	42.79	42.33	42.65
OT with fixed uniform prior	43.01	42.80	42.91	42.91
OT with dynamic prior	43.49	43.29	43.84	43.54

Dynamic-prior OT achieves 43.84% long-horizon final accuracy, outperforming uniform-prior OT by 0.93 pp and discarding routed-away samples by 0.96 pp. Moment matching and prototype contrastive alignment reach 41.51% and 42.33%, respectively. Uniform-prior OT alone yields 42.91%, close to the 42.88% discard baseline, showing that the strongest long-horizon benefit comes from transport-based repair with the adaptive marginal. The proposed treatment also gives the highest accuracy in T-CS and T-CS-LS (Table VII).

Conservative prior and centroid updates

Section III-D specifies a conservatively adaptive class marginal. Stop-gradient predictions from the full batch update an EMA initialized at the uniform distribution, with update rate $\rho = 0.01$. The OT marginal combines 99% uniform mass with 1% of this estimate ($\alpha_b = 0.01$), giving an ℓ_1 deviation from uniform of at most 0.02 and a class-wise floor of $0.99/K$. This construction limits the influence of noisy early predictions while allowing gradual adjustment to target class prevalence.

The centroids are initialized from normalized rows of the frozen source classifier and updated outside backpropagation using soft OT-weighted feature means with momentum $\rho_M = 0.99$ (Eq. (16)), followed by normalization. If the numerical transport mass of a class is negligible, its previous centroid is retained. Source initialization provides the initial semantic reference, and momentum limits the rate of prototype movement during online adaptation. Sustained bias in routed-away features can accumulate into centroid drift, which motivates the routing-ratio stress test below.

Small and dominant routed-away subsets

The revised method handles empty subsets explicitly: an empty routed-away subset sets the geometry loss to zero and leaves the centroids unchanged; an empty trusted subset skips the adaptation step and retains the model, centroids, and prior state. For small non-empty routed-away subsets, momentum $\rho_M = 0.99$ moderates each centroid update, and negligible-mass classes retain their previous centroids. Table VIII directly evaluates this regime: at a 10% routed-away proportion, DCF achieves 42.98% accuracy with centroid drift of 0.0052, showing stable geometry repair in the tested small-subset setting.

Table VIII. Routing-ratio stress test on ImageNet-C (T-CS). Controlled ratios are imposed per mini-batch using PSR-based ranking; Native uses the original routing. Acc. denotes average top-1 accuracy, and centroid drift denotes the mean ℓ_2 prototype displacement from the initial online reference.

Routed-away proportion	T-CS accuracy (%)	Centroid drift
Native routing	43.49	0.0518
10%	42.98	0.0052
25%	43.49	0.0217
50%	43.18	0.0829
75%	41.64	0.1268
90%	22.12	0.0468

With 10%–50% of samples routed away, DCF maintains 42.98%–43.49% accuracy, with centroid drift ranging from 0.0052 to 0.0829. Accuracy decreases to 41.64% at 75% and 22.12% at 90%, identifying a failure regime as the batch balance shifts sharply toward routed-away geometry repair. At 90%, centroid drift is 0.0468, while approximately 10% of each batch supplies trusted consistency supervision. The trusted loss remains normalized by subset size, with its predictive signal estimated from this smaller sample pool. Soft OT alignment and cross-view consistency constrain routed-away features, and

centroid momentum smooths prototype movement. Classification accuracy also depends on the alignment of individual features with their correct class references. The observed combination of limited prototype displacement and reduced accuracy distinguishes centroid-level stability from sample-level class separation. Together, these results establish a broad operating range for RGR and show that sustained adaptation benefits from coordinating trusted predictive supervision with routed-away geometry repair.

Changes in the manuscript: Section III-D, pp. 4–5, Eqs. (10)–(16); Algorithm 1, p. 6; Section IV-C, “Analysis of Routed-away Geometry Repair,” pp. 9–10; Tables VII–VIII, p. 10. The centroid update referred to as Eq. (19) in the original review is Eq. (16) in the revision.

R1.3: Computational overhead and lightweight deployment

DCF involves Sinkhorn iterations for OT, curvature estimation per layer, and multiple forward passes for stressed views, leading to higher training memory and runtime (Fig. 12 shows ~0.005s/image on A100). For edge deployment, this may be prohibitive. The paper should report FLOPs or memory usage in detail and discuss possible approximations (e.g., reducing Sinkhorn iterations, using cheaper curvature proxies) without sacrificing stability. A comparison with lightweight baselines (e.g., BN Adapt) in terms of throughput would contextualize the trade-off.

Online response (R1.3) ↗

PDF diff ↗

► Response:

We expanded the computational evaluation to include FLOPs, peak GPU memory, per-image runtime, and accuracy, and introduced a measured lightweight configuration, DCF-Lite.

All measurements use an NVIDIA A100, ResNet-50, and ImageNet-C under the long-horizon T-CS protocol, with batch size 64. The comparison includes BN Adapt and representative lightweight or stable baselines so that the accuracy–resource trade-off can be evaluated under a consistent setup. We report these resource metrics in Table XI below (following the table numbering in Section IV-C of the revised manuscript).

Table XI. Computational overhead and lightweight variant (reproduced from Table XI in Section IV-C of the revised manuscript). Comparison of FLOPs, peak memory, per-image runtime, and average accuracy on ImageNet-C (long-horizon T-CS, ResNet-50) measured on an NVIDIA A100.

Method	FLOPs (G)	Peak memory (MB)	Latency (ms/image)	Avg. Acc. (%)
No Adapt	4.1	1,834	0.50	7.19
BN Adapt	4.1	1,944	0.50	29.83
Tent	8.2	5,685	1.40	15.79
CoTTA	24.6	13,892	2.80	34.54
SAR	16.4	6,936	3.00	38.90
DeYO	10.0	5,895	2.10	24.18
AEA	8.3	6,936	1.60	7.60
PTTA	16.4	11,872	4.10	22.63
RoTTA	28.9	12,952	18.70	16.39
LAW	16.4	13,364	6.90	26.54
TRIBE	65.6	18,405	15.60	25.07
SPA	49.1	9,635	10.30	12.50
DCF full	24.6	14,690	5.20	43.48
DCF-Lite	9.4	6,080	3.19	42.06

Relative to BN Adapt, full DCF improves long-horizon average accuracy by 13.65 pp. At batch size 64, the effective throughputs calculated from the reported per-image runtimes are approximately 2,000 images/s for BN Adapt, 192 images/s for full DCF, and 313 images/s for DCF-Lite.

Measured approximations

The revised manuscript describes four cost-saving choices for DCF-Lite: removing persistent source/candidate model copies and per-sample gradients, retaining gradients only through the clean adaptation branch, using a detached Fourier probe for routing by DCF’s Probe-supported Sample Routing (PSR) module and for transport assignment, and reducing N_{sk} to 1.

DCF-Lite preserves the source parameter anchor as a tensor snapshot and caches a fixed curvature reference before the first optimizer update. This reference is the element-wise squared gradient of the first adaptation loss, capped element-wise at 10^{-4} , and is reused in subsequent comparisons with the current curvature proxy for retention-gate computation.

On the same A100 setup, Lite reduces FLOPs from 24.6 G to 9.4 G, peak memory from 14,690 MB to 6,080 MB, and latency from 5.20 to 3.19 ms/image, achieving 42.06% accuracy—1.42 pp below full DCF and 3.16 pp above SAR. Figure 12(d) supports low Sinkhorn iteration counts across the tested regularization range. Table IX reports accuracy gains of 0.09 and 0.22 pp for the Expected Fisher and Hutchinson proxies, respectively, with additional runtimes of 1.9 and 6.8 ms/image. The default proxy therefore offers a favorable accuracy–cost trade-off.

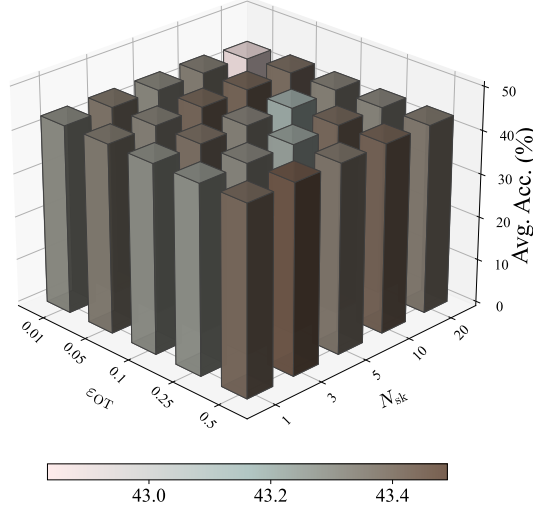


Fig. 12(d). Hyperparameter sensitivity. Stability across Sinkhorn iteration counts N_{sk} and entropic regularization values ϵ_{OT} , showing stable performance with few iterations.

These measurements establish DCF-Lite as an efficient implementation of the route–adapt–retain design, combining strong long-horizon accuracy with substantially lower computation, memory use, and runtime.

Changes in the manuscript: Section IV-C, “Computational Overhead,” pp. 11–12; Table XI, p. 12; Table IX, p. 10; Fig. 12(d), p. 12. Table XI supersedes the runtime-only comparison cited as Fig. 12 in the original review.

R1.4: Layer retention patterns and alternative proxies

CLR uses curvature mismatch to gate retention, but the paper does not analyze which layers (early vs. late) are typically retained or pulled back. Figure 8 shows drift trajectories but not per-layer retention gates μ_t^l . Providing a heatmap of μ_t^l over time and across corruption types would reveal whether the method consistently preserves low-level (e.g., first conv) or high-level (classifier) layers. Additionally, the curvature proxy is estimated from pseudo-label gradients, which themselves may be noisy; evaluating CLR with Fisher information or Hessian approximations would be insightful.

[Online response \(R1.4\)](#)

[PDF diff](#)

► Response:

We added the requested gate visualization for DCF’s Curvature-aware Layer Retention (CLR) module and comparisons with alternative curvature estimators. The revised analysis separates parameter drift, shown in Fig. 8, from the retention gate itself, shown directly in the new Fig. 9.

Which groups are retained and when

CLR acts on the trainable normalization affine parameters. In ResNet, these parameters are grouped into the stem normalization group (`initial_bn`) and four architectural stages (`layer1`–`layer4`), each containing multiple residual blocks. In ViT, grouping follows encoder blocks.

Figure 9(a) follows the gates over the long-horizon stream, and Fig. 9(b) summarizes their corruption-dependent behavior. Layer 4 remains near 0.98, while the stem and layer 1 generally lie around 0.93–0.96, layer 2 around 0.95–0.96, and layer 3

can fall to approximately 0.92 under stronger mismatch. Motion most strongly suppresses the stem gate, Glass affects layer 1, and Frost produces the strongest reduction in layer 3. For the earlier and intermediate groups, the gates recover toward 0.95–0.96 after the reductions around domain indices 30–45, reflecting a temporary increase in source anchoring followed by renewed support for candidate updates.

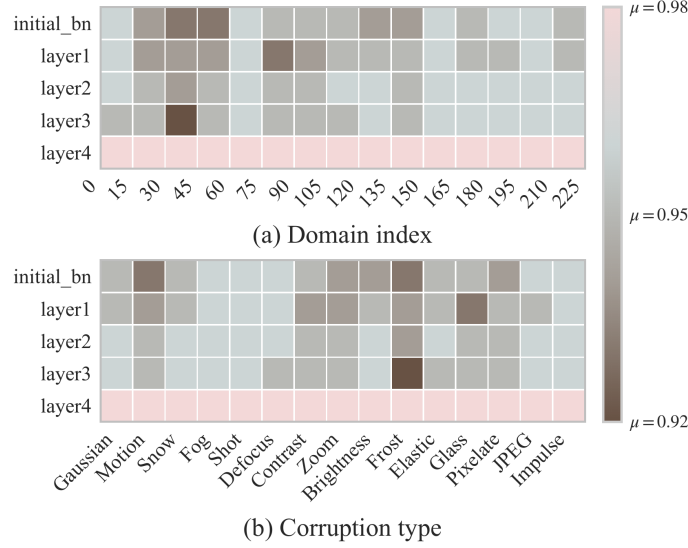


Fig. 9. Per-layer retention gates μ_t^l of CLR (ResNet-50). (a) Gates over the long-horizon stream (Domain index 0 → 225). (b) Gates per corruption type on T-CS.

A larger gate favors the candidate parameters, and a smaller gate strengthens source anchoring. In Eq. (18), the effective candidate weight for coordinate j in group l is $a_{t,j}^l = \mu_t^l \tilde{I}_{t,j}^l / [\mu_t^l \tilde{I}_{t,j}^l + (1 - \mu_t^l) \tilde{I}_{0,j}^l]$. Figure 9 visualizes the group-wise gate μ_t^l , and the actual retention update combines this gate with the coordinate-wise proxy values.

Granularity and curvature proxy comparisons

Table IX compares control granularities and curvature proxies under the long-horizon T-CS protocol. Table numbering follows the revised manuscript.

Table IX. Ablation of design choices in CLR on ImageNet-C (reproduced from Table IX in Section IV-C of the revised manuscript). We compare different control granularities and curvature-aware proxy estimators.

Design choice	Reported accuracy (%)	Extra latency (ms/image)
<i>(a) Control granularity</i>		
Per-normalization-layer control	42.92	0
Per-stage control (default)	43.48	0
<i>(b) Curvature-aware proxy estimator</i>		
Per-sample gradient-square proxy (default)	43.48	0
Expected Fisher proxy	43.57	1.9
Hutchinson-based diagonal approximation	43.70	6.8

As shown in Table IX, stage-level control outperforms normalization-layer control by 0.56 pp. Crucially, regarding the curvature estimator, more sophisticated proxies yield only marginal accuracy gains (+0.09 pp for Expected Fisher and +0.22 pp for Hutchinson) while incurring substantial latency overheads (+1.9 and +6.8 ms/image). These results show that the default per-sample gradient-square proxy offers a favorable accuracy–cost trade-off among the evaluated proxies.

Implementation details of alternative proxies.: To directly address the concern regarding potential gradient noise from hard pseudo-labels, we formulated and evaluated these two higher-fidelity alternatives as follows:

- **Expected Fisher proxy:** Instead of taking gradients at the hard pseudo-label $\arg \max$, this proxy marginalizes over the model’s predictive posterior $p_{\theta_t^+}(k | x_{t,i})$ to dampen misclassification noise:

$$I_{t,\text{EF}}^l = \frac{1}{N_t} \sum_{i=1}^{N_t} \sum_{k=1}^K p_{\theta_t^+}(k | x_{t,i}) \left[\nabla_{\theta_t^+, l} \log p_{\theta_t^+}(k | x_{t,i}) \right]^{\odot 2},$$

with the source reference $I_{0,EF}^l$ evaluated symmetrically at θ_0 on the same batch.

- **Hutchinson-based diagonal Hessian:** Probing the diagonal of the true loss Hessian $H_t^l = \nabla_{\theta^l}^2 \mathcal{L}_{TTA}$ via randomized trace estimation:

$$I_{t,Hutch}^l = \left[\frac{1}{M} \sum_{m=1}^M v_m \odot (H_t^l v_m) \right]_+,$$

where $v_m \in \{-1, +1\}^{d_l}$ are i.i.d. Rademacher vectors ($M = 2$), and $[\cdot]_+$ enforces non-negativity for the proximal objective in Eq. (17). Each Hessian-vector product $H_t^l v_m = \nabla_{\theta^l} \langle \nabla_{\theta^l} \mathcal{L}_{TTA}, v_m \rangle$ is computed matrix-free via Pearlmutter’s algorithm with one additional backward pass.

Changes in the manuscript: Section III-E, pp. 5–6, Eqs. (17)–(20); Section IV-A, “Models and Implementation Details,” p. 6; Section IV-C, “Analysis of Curvature-aware Layer Retention,” Fig. 9 and Table IX, p. 10.

R1.5: Additional literature on uncertainty-aware modeling

The literature review part can be improved by including more recent publications on machine learning methods, which can refer to Uncertainty-aware, high-precision multi-step prediction of structural health monitoring sensor streams under extreme typhoon events: an enhanced Bayesian dynamic linear model leveraging the kernel regression basis function for severe environmental adaptation, Bayesian Network in Structural Health Monitoring: Theoretical Background and Applications Review, Towards high-accuracy data modelling, uncertainty quantification and correlation analysis for SHM measurements during typhoon events using an improved most likely heteroscedastic Gaussian process.

Online response (R1.5) [↗](#)

PDF diff [↗](#)

► Response:

Thank you for suggesting these studies. We have incorporated all three references into Section II as cross-disciplinary work on uncertainty-aware modeling under evolving environmental conditions. The added references are:

[19] Q.-A. Wang et al., “Uncertainty-Aware, High-Precision Multi-Step Prediction of Structural Health Monitoring Sensor Streams under Extreme Typhoon Events: An Enhanced Bayesian Dynamic Linear Model Leveraging the Kernel Regression Basis Function for Severe Environmental Adaptation,” *Measurement*, vol. 262, article 120050, 2026.

[20] Q.-A. Wang, A.-W. Lu, Y.-Q. Ni, J.-F. Wang, and Z.-G. Ma, “Bayesian Network in Structural Health Monitoring: Theoretical Background and Applications Review,” *Sensors*, vol. 25, no. 12, article 3577, 2025.

[21] Q.-A. Wang et al., “Towards High-Accuracy Data Modelling, Uncertainty Quantification and Correlation Analysis for SHM Measurements during Typhoon Events Using an Improved Most Likely Heteroscedastic Gaussian Process,” *Smart Structures and Systems*, vol. 32, no. 4, pp. 267–279, 2023.

These studies contribute Bayesian dynamic modeling, Bayesian-network reasoning, and heteroscedastic uncertainty quantification for non-stationary structural-health-monitoring streams. We added them to broaden the discussion of uncertainty under evolving environments. DCF addresses online visual adaptation from unlabeled, temporally correlated observations, with a focus on coordinating evidence routing, geometry repair, and layer-wise update retention.

Changes in the manuscript: Section II, “Adaptation without Target Data,” p. 3; references [19]–[21], p. 13.

R1.6: Variability, significance, and transfer beyond ImageNet

In Figure 10, DCF shows positive gains across all source domains, but the paper does not report the variance or statistical significance. Some cells show marginal improvements (<1%), which may not be robust. The authors should include standard deviations over multiple runs and perform a significance test (e.g., paired t-test) against the best baseline. Furthermore, the cross-domain evaluation adapts on one corruption and tests on the remaining 14, but the test set still shares the same ImageNet classes; evaluating on a completely different dataset (e.g., CIFAR-10-C or a medical imaging shift) would better demonstrate generalizable representation preservation. Adding such an experiment or discussing the limitation would strengthen the paper’s claims.

Online response (R1.6) [↗](#)

PDF diff [↗](#)

► Response:

We agree that the cross-domain results should be supported by run-to-run variability and formal significance testing. We therefore added five matched runs, paired significance tests, and a second transfer benchmark on DomainNet-126. The resulting analysis shows that DCF’s aggregate transfer advantage is statistically significant on both ImageNet-C and DomainNet-126, while the per-transfer mean $\pm$ standard deviation values characterize how those gains are distributed across individual transfer settings.

Variability across five runs and paired comparisons

The revised Fig. 11 replaces the original Fig. 10. Panel (a) updates the existing ImageNet-C analysis with five-run variability, while panel (b) adds the new DomainNet-126 benchmark. Both panels report mean $\pm$ standard deviation over five matched runs. On ImageNet-C, DCF achieves a mean gain over No Adapt of 30.30 ± 0.39 pp, compared with 27.95 ± 0.05 pp for AEA, the strongest aggregate baseline in this comparison.

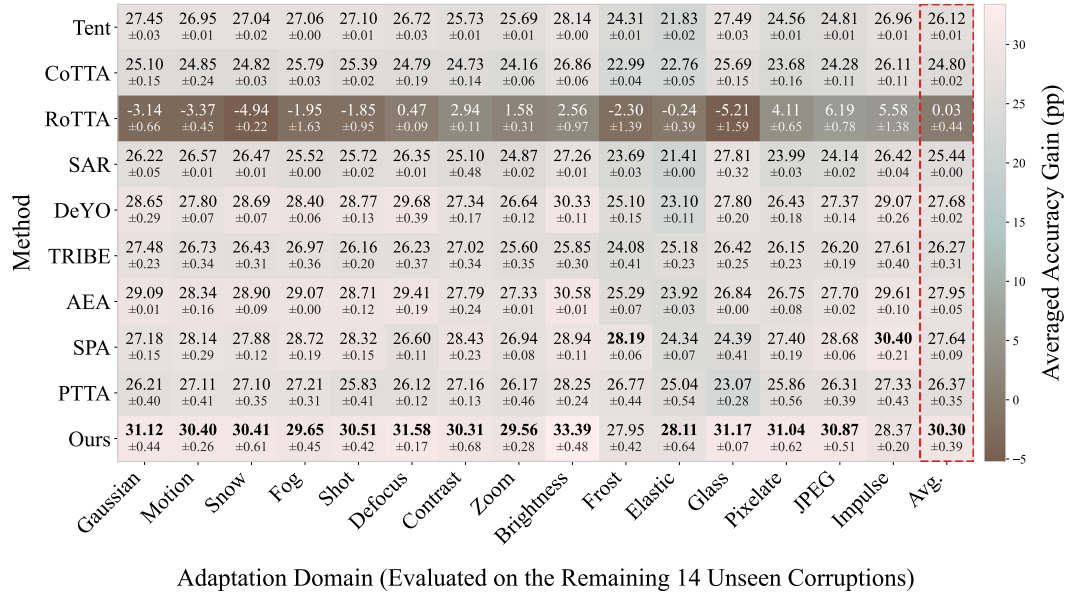


Fig. 11(a). Cross-domain transfer. ImageNet-C accuracy gains (in percentage points) over No Adapt, averaged across all unseen corruption domains. Results are reported as mean $\pm$ standard deviation over five runs.

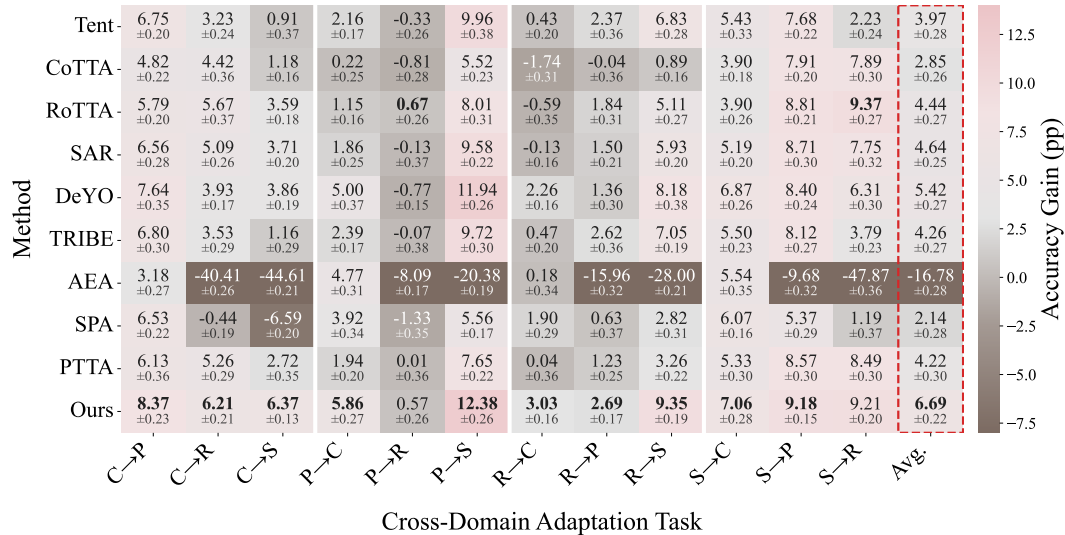


Fig. 11(b). Cross-domain transfer. DomainNet-126 accuracy gains over No Adapt for each cross-domain transfer pair (Adaptation $\rightarrow$ Evaluation, where C: Clipart, P: Painting, R: Real, and S: Sketch). Results are reported as mean $\pm$ standard deviation over five runs.

Summary of cross-domain statistical results reported in revised Section IV-C

Evaluation	DCF	Best aggregate baseline	Difference	Paired test
ImageNet-C gain over No Adapt	30.30 ± 0.39 pp	AEA / 27.95 ± 0.05 pp	+2.35 pp	$t(4) = 14.30; p = 1.39 \times 10^{-4}$
DomainNet-126 gain over No Adapt	6.69 ± 0.22 pp	DeYO / 5.42 ± 0.27 pp	+1.27 pp	$t(4) = 12.01; p = 2.75 \times 10^{-4}$

The paired tests compare five matched run-level cross-domain aggregates for DCF and each benchmark’s strongest aggregate baseline. This analysis establishes statistically significant overall gains over second-best AEA on ImageNet-C and DeYO on DomainNet-126. Per-transfer means and standard deviations characterize the magnitude and run-to-run variability of gains across individual adaptation domains and evaluation pairs.

Fig. 11(a): Domain-aggregated transfer on ImageNet-C. Each entry reports transfer gain after adaptation to one of the 15 corruption domains, averaged across the remaining 14 unseen corruptions without further updates. DCF exceeds the domain-wise strongest baseline in 13 of 15 domains, with advantages above 1.5 pp in 12 domains and a maximum of 3.64 pp. The domain-specific margins are +0.58 pp on Fog, -0.24 pp on Frost, and -2.03 pp on Impulse relative to the strongest baseline in each domain. These domain-wise margins map transfer gains across corruptions, and the five matched run-level aggregate comparisons quantify DCF’s statistically significant overall advantage.

Fig. 11(b): Pairwise transfer on DomainNet-126. Each entry reports mean $\pm$ standard deviation over five runs for a specific Adaptation $\rightarrow$ Evaluation pair. DCF exceeds DeYO in all 12 pairs and yields positive gains over No Adapt throughout this matrix (0.57–12.38 pp). It achieves the highest mean gain in 10 of 12 pairs; the exceptions are P $\rightarrow$ R (0.57 versus RoTTA’s 0.67 pp) and S $\rightarrow$ R (9.21 versus RoTTA’s 9.37 pp). The positive gains across all 12 pairs and the significant aggregate advantage over DeYO demonstrate that DCF sustains transfer performance across the DomainNet visual-style shifts.

Transfer on a distinct dataset

To test transfer beyond ImageNet corruptions, we added **DomainNet-126** [46], [47], which contains 126 classes across Clipart, Painting, Real, and Sketch. This benchmark complements ImageNet-C by evaluating cross-domain transfer across substantial changes in visual style on a separate dataset.

For fair comparison and standardized evaluation, we directly adopt the unified source-domain ResNet-50 checkpoints provided by MME [46]. On DomainNet-126, models adapt on one domain and are evaluated on the remaining domains without further parameter updates. DCF achieves positive gains over No Adapt across all 12 transfer pairs (+0.57 to +12.38 pp) and exceeds DeYO in every pair, with a significant aggregate accuracy advantage (61.40% versus 60.13%, $p = 2.75 \times 10^{-4}$). This second benchmark extends the transfer evaluation to a separate 126-class recognition task with substantial changes in visual style, strengthening the evidence that DCF maintains generalization across adaptation domains.

We note that our cross-domain evaluations focus on closed-set visual recognition under covariate and style shifts; extending DCF to open-set adaptation or other imaging modalities or application domains remains a valuable avenue for future work.

Changes in the manuscript: Section IV-A, “Benchmarks,” p. 6; Section IV-C, “Cross-Domain Transfer,” and Fig. 11(a)–(b), p. 11.

Response to Reviewer 2

Thank you for the positive assessment of the route–adapt–retain framework and the long-horizon evidence. We have addressed the reproducibility, visualization, practical-guidance, and literature suggestions as follows.

Reviewer 2: Comment Overview and Navigation

Item	Comment Title	Key Revision	Direct Jump
R2.1	Trainable parameters and the meaning of a layer	Tent affine setup; ResNet/ViT block grouping	Jump → p. 19
R2.2	Explicit definitions of the four routing regions	Mathematical inequalities for Areas 1–4	Jump → p. 19
R2.3	Practical threshold guidance and OT settings	Default threshold guidance and OT specifications	Jump → p. 20
R2.4	Recent work on continual adaptation	Integrated GOLD and recent continual-TTA survey	Jump → p. 21

R2.1: Trainable parameters and the meaning of a layer

The implementation section states that the experiments follow the official Tent setup. For clarity and reproducibility, please explicitly specify which model parameters are optimized during TTA (e.g., all network parameters, only the affine parameters of normalization layers, or another subset). It would also be helpful to clarify how a layer is defined when applying Curvature-aware Layer Retention. For convolutional architectures, does this refer to an individual normalization layer, a residual block, or a larger network stage?

[Online response \(R2.1\)](#)

[PDF diff](#)

► Response:

We now specify which parameters are optimized and how they are grouped for DCF’s Curvature-aware Layer Retention (CLR) module. Following Tent, SGD optimizes the normalization scale and bias parameters (γ, β) , using a learning rate of 2.5×10^{-4} and momentum 0.9. The convolution kernels and classifier weights are fixed during adaptation.

For ResNet, CLR groups the trainable affine parameters into the stem normalization group (`initial_bn`) and four architectural stages (`layer1–layer4`). For ViT, grouping follows encoder blocks. These groups define the rows in the retention-gate visualization.

We additionally compare stage-level grouping against per-normalization-layer control for ResNet. Table IX reports 43.48% and 42.92% accuracy, respectively, supporting the default stage-level grouping.

Changes in the manuscript: Section IV-A, “Models and Implementation Details,” p. 6; Section III-A, p. 3; Fig. 9 and Table IX, p. 10.

R2.2: Explicit definitions of the four routing regions

The PCS–entropy visualization in Fig. 7 is a useful addition and provides good intuition for the routing mechanism. The caption already identifies Area 1 as the low-entropy/high-PCS trusted region, but the exact conditions defining all four regions are not stated explicitly. I suggest defining Areas 1–4 and their corresponding entropy/PCS threshold conditions directly in the figure caption or in the accompanying text. This would allow readers to interpret the visualization without repeatedly referring back to the routing equations.

[Online response \(R2.2\)](#)

[PDF diff](#)

► **Response:**

Section III-C defines Areas 1–4 for DCF’s Probe-supported Sample Routing (PSR) module after Eqs. (6)–(7) using the routing thresholds v_{Ent} and v_{PCS} . For ImageNet-C ($K = 1000$), their defaults are $0.6 \ln 1000 \approx 4.145$ and 0.2, respectively. Figure 7 uses $E = 3$ and $s = 0.5$ to visualize the diagnostic regions. The table below specifies the routing conditions, with E denoting predictive entropy and s denoting the Perturbation Consistency Score (PCS).

Definitions added to revised Section III-C

Region	Entropy condition	PCS condition	Role
Area 1	$E < v_{\text{Ent}}$	$s > v_{\text{PCS}}$	Trusted
Area 2	$E \geq v_{\text{Ent}}$	$s > v_{\text{PCS}}$	Routed away
Area 3	$E < v_{\text{Ent}}$	$s \leq v_{\text{PCS}}$	Routed away
Area 4	$E \geq v_{\text{Ent}}$	$s \leq v_{\text{PCS}}$	Routed away

The routing rule assigns Area 1 to trusted consistency supervision and Areas 2–4 to the routed-away set used for geometry repair. The explicit inequalities also highlight the role of Area 3: these samples appear confident according to entropy but exhibit insufficient stress responsiveness, precisely the regime in which entropy-only selection can retain stress-inert predictions.

Changes in the manuscript: Section III-C, p. 4, following Eqs. (6)–(7); Fig. 7, p. 9, provides the corresponding PCS–entropy diagnostic visualization.

R2.3: Practical threshold guidance and OT settings

The sensitivity analysis suggests that DCF remains effective across a reasonable range of routing hyperparameters. It would be useful to provide brief practical guidance, or a simple rule of thumb, for selecting the entropy and PCS thresholds when applying DCF to a new dataset. In addition, for reproducibility, please consider reporting the remaining implementation details of the optimal transport procedure, such as the number of Sinkhorn iterations and other relevant solver settings.

[Online response \(R2.3\)](#)
[PDF diff](#)
► **Response:**

We now provide a starting rule for the thresholds of DCF’s Probe-supported Sample Routing (PSR) module and specify the OT solver settings for DCF’s Routed-away Geometry Repair (RGR) module. For a new K -class dataset, we recommend starting from $v_{\text{Ent}} = 0.6 \ln K$ and $v_{\text{PCS}} = 0.2$ with the specified Fourier probe and $\lambda = 0.2$. The entropy threshold scales with the maximum class entropy, whereas the Perturbation Consistency Score (PCS) threshold is interpreted jointly with the probe strength and the model probability response. Figures 12(b)–(c) support these settings as practical starting values. For OT, the default is Sinkhorn–Knopp with $N_{\text{sk}} = 3$ iterations and $\varepsilon_{\text{OT}} = 0.5$; Fig. 12(d) evaluates alternative iteration counts and regularization values.

Figure 12(b) shows a broad high-accuracy region around this default setting, supporting it as a robust practical initialization. The routing-ratio analysis in Table VIII further explains the trade-off between trusted-set selectivity and the amount of reliable supervision available to the online update.

Implementation settings specified in the revised manuscript

OT or routing quantity	Value or specification
Sinkhorn–Knopp iterations	$N_{\text{sk}} = 3$
Entropic regularization	$\varepsilon_{\text{OT}} = 0.5$
Transport cost	1 – cosine similarity of normalized features and centroids
Sample marginal	Uniform over the routed-away subset
Class marginal	EMA prediction prior mixed with the uniform prior
Prior EMA and uniform mixing	$\rho = 0.01$; $\alpha_b = 0.01$; uniform initialization
Centroid momentum	$\rho_M = 0.99$, followed by normalization
Gradient treatment	Transport plans and prior predictions are detached
Routing defaults	$v_{\text{PCS}} = 0.2$; $v_{\text{Ent}} = 0.6 \ln K$
Fourier amplitude default	$\lambda = 0.2$

Section III-D further specifies the shared centroids, two-view transport plans, cross-view geometry objective, empty-subset safeguards, and negligible-class-mass handling. Figure 12(d) jointly studies Sinkhorn iterations and entropic regularization. The stable adaptation accuracy in Fig. 12(d) supports using a small, fixed number of Sinkhorn iterations for online geometry repair. See Response R1.3 (p. 13) for Fig. 12(d).

Changes in the manuscript: Section IV-A, “Models and Implementation Details,” p. 6; Section III-D, pp. 4–5, Eqs. (10)–(16); “Hyperparameter Sensitivity,” p. 11, and Fig. 12(b), (d), p. 12.

R2.4: Recent work on continual adaptation

Given the rapid development of continual and long-horizon TTA, the related-work discussion could potentially be strengthened by briefly acknowledging very recent studies on long-horizon stability, where relevant. For example:
 [1] *The Golden Subspace: Where Efficiency Meets Generalization in Continual Test-Time Adaptation*, CVPR 2026.
 [2] *Continual Test-Time Adaptation in Computer Vision: Methods, Benchmarks, and Future Directions*, TMLR 2026.

 Online response (R2.4) 

 PDF diff 

► Response:

Section II now cites both suggested 2026 studies and explains how their approaches relate to DCF.

[38] G. Lai, D.-W. Zhou, Z. Li, and H.-J. Ye, “The Golden Subspace: Where Efficiency Meets Generalization in Continual Test-Time Adaptation,” CVPR, 2026.

GOLD constrains updates to a classifier-aligned low-rank subspace. DCF takes a complementary approach by controlling adaptation evidence, reusing routed-away geometry, and selectively retaining candidate updates across layer groups according to source mismatch.

[28] S. K. Maharana et al., “Continual Test-Time Adaptation in Computer Vision: Methods, Benchmarks, and Future Directions,” Transactions on Machine Learning Research, 2026.

The survey provides context for teacher models, memory, reset strategies, and long-horizon stability. DCF specifically links heterogeneous target evidence to layer-wise update retention.

Changes in the manuscript: Section II, “Test-Time adaptation” and “Reliability- and Stability-aware TTA,” p. 3; references [28] and [38], p. 13.